Trajectory-Aligned Latent Handoff for Efficient High-Resolution Video Diffusion

Anonymous submission

Abstract

Despite recent advances, high-resolution video diffusion models suffer from prohibitive inference latency due to their iterative denoising computations over long spatiotemporal latent sequences, severely hindering real-time content generation and interactive applications. While the early stages of diffusion sampling primarily establish global scene layout and macro-motion dynamics, the later stages progressively inject fine high-frequency details. This functional separation opens a valuable window for mixed-resolution inference. However, dynamic resolution handoff is far from a simple tensor resizing operation; it is inherently challenged by the compound effects of residual denoising gaps, cross-scale representation biases of video VAEs, and high-resolution trajectory mismatch. In this paper, we propose **Trajectory-Aligned Latent Handoff (TALH)**, a lightweight latent-space framework tailored for efficient high-resolution video generation. TALH strategically shifts the bulk of denoising computations to a low-resolution latent space and decouples the handoff process into three complementary components: a *Trajectory Alignment Adapter (TAA)* that leverages LoRA parameters to calibrate the residual trajectory gap at the switching step; a *Clean Latent Lifter (CLL)* that maps the cross-resolution clean latents; and a *High-Resolution Trajectory Re-entry (HTR)* protocol that seamlessly injects the lifted latents back into the high-resolution denoising trajectory of a frozen generator. Operating under a model-endogenous self-distillation paradigm, both TAA and CLL are supervised directly by the target generator, entirely eliminating the need for external paired videos, pre-trained upscalers, or heavy teacher models. Evaluated on a 50-step Wan2.1 model for $368 \times 640 \rightarrow 720 \times 1248$ resolution handoff, our quality-oriented variant (TALH-Q) and efficiency-oriented variant (TALH-E) achieve $1.83\times$ and $2.22\times$ end-to-end speedups, respectively, while preserving 97.76% and 97.53% of native high-resolution sampling performance on VBench-5. Component-wise analysis demonstrates that for $480 \times 832 \rightarrow 720 \times 1248$ upscaling, CLL reduces the latent L_1 error by 48.6% and LPIPS by 73.3% compared to trilinear interpolation. Furthermore, on the 368×640 low-resolution trajectory, TAA reduces the endpoint L_1 error at steps 40 and 45 by 25.8% and 21.0%, respectively. TALH consistently exhibits a robust quality-efficiency Pareto frontier across both standard 50-step and advanced 4-step distilled models.

1 Introduction

Latent video diffusion models, exemplified by recent architectures like Wan, have demonstrated remarkable capabilities in generating videos with strong semantic consistency, vivid motion, and intricate visual details (Wan et al. 2025). However, their inference cost scales aggressively with spatial resolution, video length, and temporal sampling steps. For Diffusion Transformers (DiTs), each denoising iteration requires processing the full sequence of spatiotemporal tokens. When all sampling steps are executed at the target high resolution, the early stages—which only model coarse structures and macro-motions—are forced to bear the full burden of high-resolution computation. This property significantly exacerbates the response latency for online generation, severely hindering the deployment of high-resolution video models in interactive content creation and real-time services.

Diffusion trajectories inherently partition generative tasks across different temporal stages: early steps dictate the primary structure, scene layout, and motion patterns, while later steps progressively add local textures and high-frequency details. Consequently, a promising acceleration paradigm is to perform global content modeling in a low-resolution latent space first, and then hand off to the target high resolution in the later sampling phase, using only a few high-resolution steps to restore details. For instance, LightX2V introduces a changing-resolution inference that employs trilinear interpolation for dynamic-resolution sampling (ModelTC 2026), while MrFlow uncovers the efficiency benefits of low-resolution structure generation coupled with low-noise high-resolution refinement for image flow matching models (Zheng et al. 2026). Since these methods alter the spatial computational scale of individual sampling stages, they are fundamentally orthogonal to and complementary with distillation methods that reduce temporal sampling steps.

Nevertheless, dynamic resolution handoff is far more complex than simple tensor scaling. A substantial *residual denoising gap* persists between the single-step clean prediction at step s and the endpoint of the complete low-resolution trajectory, a gap that becomes increasingly pronounced at earlier handoff steps. Concurrently, the latent representations of video VAEs do not satisfy strict scale equivariance; standard interpolation merely resizes the latent grid but fails to capture the intricate cross-resolution correspondences jointly determined by the VAE architecture, video content, and

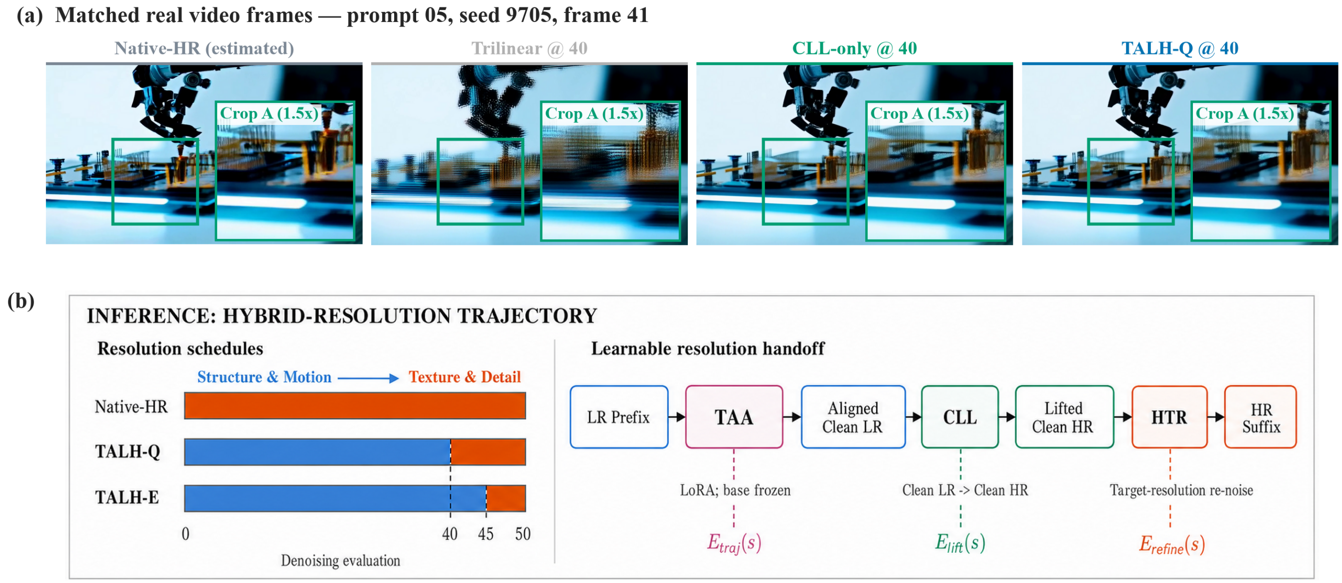

Figure 1: Real-frame comparison and acceleration mechanism of TALH. (a) Full frames and identical $1.5\times$-magnified Crop A regions compare the content-aligned Native-HR (estimated) trajectory with Trilinear, CLL-only, and TALH-Q outputs for prompt 05, seed 9705, frame 41. The estimated trajectory is a complete 368p sampling path; the handoff outputs are 720p. (b) Native-HR Sampling executes all 50 denoising evaluations at high resolution, whereas TALH-Q and TALH-E move the first 40 and 45 evaluations to the low-resolution latent space and retain 10 and 5 high-resolution suffix evaluations, respectively.

scale transitions. Finally, the lifted latents must re-enter the high-resolution sampling trajectory so that the frozen generator can continue injecting textures and correcting upscaling residuals. These three distinct error sources—trajectory gap, scale discrepancy, and refinement residual—collectively dictate the final quality of mixed-resolution sampling.

Directly mapping the low-resolution prediction at the handoff step to the high-resolution endpoint forces a single Joint Trajectory-Scale Lifter (JTSL) to simultaneously resolve trajectory calibration, scale transformation, and texture inference. Because the residual trajectory gap fluctuates with time steps, schedulers, and text prompts, and since high-resolution details are underdetermined by low-resolution inputs, such joint regression heavily suffers from high-frequency attenuation. Qualitative results from our handoff-step scanning experiments corroborate this, showing that JTSL introduces severe blurring compared to a pure cross-scale mapping between clean latents. This crucial observation motivates us to decompose dynamic resolution handoff into well-structured, localized sub-problems.

Driven by these insights, we present **TALH** (Trajectory-Aligned Latent Handoff), which decomposes dynamic resolution handoff into three modular components: TAA, CLL, and HTR. Activated exclusively at the designated handoff step, the **Trajectory Alignment Adapter (TAA)** uses LoRA to reduce the discrepancy between the current clean prediction and the complete low-resolution trajectory endpoint. The **Clean Latent Lifter (CLL)** then learns a cross-scale mapping between separately encoded low- and high-resolution clean latents. Finally, **High-Resolution Trajectory Re-entry (HTR)** re-noises the output according to the target-resolution schedule, allowing the frozen generator to complete the high-resolution suffix. Both TAA and CLL are supervised by the *same* frozen generative model: TAA uses the full low-resolution trajectory to construct trajectory alignment pairs, while CLL uses model-generated high-resolution videos and their downsampled counterparts to construct cross-resolution lifting pairs. This design yields a unified, model-endogenous self-distillation framework. Figures 1 and 2 summarize the real-frame behavior, mixed-resolution inference path, and dual supervision.

Our primary contributions are summarized as follows:

- **We propose TALH**, a lightweight latent-space framework for efficient high-resolution video generation that strategically shifts the bulk of denoising computations to a low-resolution space, restricting high-resolution processing to the sampling suffix. Its variants, TALH-Q and TALH-E, establish quality- and efficiency-oriented Pareto operating points, offering a controllable speed-quality trade-off for online inference.
- **We decouple dynamic resolution handoff** into three explicit phases—residual trajectory alignment, cross-resolution latent lifting, and generative high-resolution refinement—by designing TAA, CLL, and HTR. This mathematical decomposition significantly mitigates the epistemic uncertainty inherent in joint trajectory-scale mapping, enabling isolated diagnostics of each component.
- **We construct a model-endogenous training paradigm** that requires no external paired videos, external upscaler weights, or additional teacher models, and establish the

consistency condition for cached-prefix states. Experiments on standard 50-step Wan2.1 and 4-step distilled Wan validate TALH's system efficiency, module-level effects, and compatibility with different sampling schedules.

2 Related Work

2.1 Latent Video Generation and Cascaded Upscaling

Latent Diffusion Models transfer the generative process to a compressed latent space, providing a highly scalable foundation for high-resolution image and video synthesis (Rombach et al. 2022). Early frameworks like Video LDM, Imagen Video, and LaVie adopted spatial or temporal cascades to progressively generate high-resolution videos from low-resolution initial outputs (Blattmann et al. 2023; Ho et al. 2022; Wang et al. 2023b). More recently, architectures such as LTX-Video and LTX-2 have co-designed video VAEs, latent representations, and generative Transformers, with LTX-2 introducing an independent spatial latent upsampler (HaCohen et al. 2024, 2026). Other recent works like SimpleGVR and LUVE similarly explore cascaded pipelines consisting of a low-resolution generator followed by latent or video refinement modules (Xie et al. 2025; Zhao et al. 2026).

While these approaches typically treat the upscaler as an isolated post-processing stage after the core generation, TALH natively embeds the Clean Latent Lifter (CLL) into the original Wan sampling trajectory. By linking the lifted latents to subsequent scheduling steps via High-Resolution Trajectory Re-entry (HTR), cross-resolution scaling and high-resolution generative priors are unified within a single, continuous inference process.

2.2 Video Super-Resolution and Temporal Consistency

Advanced methods such as Upscale-A-Video, SATeCo, MGLD, and VideoGigaGAN leverage diffusion priors, spatiotemporal adapters, motion guidance, and generative restoration to elevate video resolution while suppressing flickering artifacts (Zhou et al. 2024; Chen et al. 2024; Yang et al. 2024; Xu et al. 2024). However, these approaches are primarily engineered for the restoration of already generated or real low-resolution RGB videos. In contrast, TALH operates directly within the generative latent space prior to VAE decoding. Because its upscaling outputs must participate in subsequent diffusion denoising steps, TALH targets resolution transitions inside the generation trajectory rather than independent RGB video post-processing.

2.3 Mixed-Resolution Diffusion Inference

The changing-resolution inference in LightX2V utilizes fixed interpolation to dynamically adjust latent sizes mid-sampling (ModelTC 2026). Concurrently, RALU demonstrates that training-free latent upscaling is frequently plagued by high-frequency aliasing and a resolution-dependent mismatch between noise and time steps, and subsequently designs a mixed-resolution processing scheme tailored for Diffusion Transformers (Jeong et al. 2025). For image flow matching models, MrFlow introduces a staged sampling strategy that combines low-resolution structure generation, pixel-domain super-resolution, low-intensity re-noising, and high-resolution detail refinement, demonstrating significant end-to-end acceleration and compatibility with temporal step distillation (Zheng et al. 2026). TALH extends this paradigm to video generation trajectories by learning cross-resolution lifting directly within the target model's latent space, while explicitly rectifying low-resolution tail-trajectory errors prior to the handoff.

2.4 Few-Step Distillation and On-Policy Trajectory Learning

To accelerate diffusion models, methods like Progressive Distillation, Consistency Models, LCM, and Distribution Matching Distillation minimize the required temporal sampling steps via trajectory regression, consistency mapping, or distribution matching (Salimans and Ho 2022; Song et al. 2023; Luo et al. 2023; Yin et al. 2024b,a). Frameworks such as VideoLCM and the Motion Consistency Model successfully extend these formulations to the video domain (Wang et al. 2023a; Zhai et al. 2024). Recent works like Self-Forcing, D-OPSD, and AnyFlow further emphasize enforcing strict consistency between the training states and the actual inference states of student models (Huang et al. 2025; Jiang et al. 2026; Gu et al. 2026).

TALH retains the original sampler and relies on TAA to calibrate a single resolution handoff step. Since TAA remains inactive prior to the handoff step, its parameter updates do not alter the low-resolution prefix trajectory. Leveraging this property, Section 3.5 establishes the consistency conditions that allow cached prefix states to be directly accessed during inference without training-inference distribution drift.

3 Method

3.1 Problem Definition and Overall Framework

Let $v_\phi(x_s, s, c)$ denote a frozen Wan flow predictor, where x_s represents the intermediate latent before the s-th denoising calculation, c is the text condition, and T is the total number of denoising steps. We use the superscripts L and H to signify the low-resolution and high-resolution latent spaces, respectively. Let z_T^L denote the endpoint of the complete low-resolution trajectory generated by the frozen model under a given text prompt, random seed, initial noise, and low-resolution scheduler. We define (z_0^L, z_0^H) as the cross-resolution clean latent pair obtained by encoding the low- and high-resolution versions of the same video through the Wan VAE.

TALH introduces two independently trained modules: the Trajectory Alignment Adapter (TAA) parameterized by LoRA weights $\Delta\theta_s$, and the Clean Latent Lifter (CLL) denoted as U_ψ^{CLL} with parameters ψ. High-Resolution Trajectory Re-entry (HTR) is a parameter-free re-routing operation. As illustrated in the upper panel of Figure 2, the complete data flow is formulated as:

$$x_s^L \xrightarrow{\mathrm{TAA}:v_{\phi+\Delta\theta_s}} \tilde{z}_s^L \xrightarrow{U_\psi^{\mathrm{CLL}}} \hat{z}_s^H \xrightarrow{\mathrm{HTR}} x_{s+1}^H.$$

Crucially, TAA is dedicated solely to aligning the current low-resolution clean prediction with the complete trajectory's low-resolution endpoint, while CLL focuses exclusively on the cross-resolution mapping of clean latents. These two modules are trained separately without joint backpropagation. This decoupled training strategy prevents the massive memory overhead of hosting both the 14B denoiser and the CLL module simultaneously in the compute graph, while enabling an isolated experimental analysis of trajectory and scale errors.

The handoff step simultaneously governs three distinct types of errors. Let $E_{\mathrm{traj}}(s)$ denote the residual trajectory gap before the handoff, $E_{\mathrm{lift}}(s)$ be the cross-resolution lifting error, and $E_{\mathrm{refine}}(s)$ represent the remaining high-resolution error left uncorrected after HTR. The total handoff error can be conceptualized as:

$$E_{\mathrm{handoff}}(s) \approx E_{\mathrm{traj}}(s) + E_{\mathrm{lift}}(s) + E_{\mathrm{refine}}(s).$$

This conceptual decomposition reveals the central trade-off across choices of s. A later handoff reduces the residual trajectory gap $E_{\mathrm{traj}}(s)$ and moves the CLL input closer to its clean-latent training distribution, thereby reducing $E_{\mathrm{lift}}(s)$. However, a later handoff also leaves fewer high-resolution suffix evaluations and can therefore increase the uncorrected refinement error $E_{\mathrm{refine}}(s)$. TAA, CLL, and HTR are designed to address these three error sources separately.

3.2 Model-Endogenous Dual-Supervision Data

TALH operates entirely without external paired videos, pre-trained super-resolution weights, or additional teacher models. The supervision for both CLL and TAA is generated natively by the frozen target generator, tracking two distinct conditional distributions as shown in the lower panel of Figure 2.

Cross-Resolution Lifting Pairs. We run the frozen Wan model using a set of predefined text prompts and fixed random seeds to generate high-resolution reference videos. These videos are spatially downsampled in the RGB pixel space. Both the high- and low-resolution variants are then encoded using the same Wan VAE to produce perfectly aligned cross-resolution clean latent pairs (z_0^L, z_0^H). This construction preserves the frame count, semantic content, and motion trajectories, forcing CLL to learn cross-scale correspondences tightly bound to the target model's specific VAE and generative distribution.

Trajectory Alignment Pairs. Under identical text prompts, random seeds, initial noise, and schedulers, we roll out the complete low-resolution trajectory of the frozen model. We cache the intermediate low-resolution state x_s^L immediately preceding the handoff step and utilize the final low-resolution trajectory endpoint z_T^L as the ground-truth target. Consequently, CLL receives cross-resolution structural supervision, while TAA receives cross-timestep trajectory supervision, forming a unified model-endogenous resolution-trajectory self-distillation framework.

Although both types of supervision share the same base generative model, they map to different conditional distributions: the low-resolution input for CLL is derived from downsampled high-resolution video encodings, whereas TAA directly targets the endpoints of native low-resolution trajectories. Section 4.3 evaluates their end-to-end synergy via a 2×2 factorial design crossing trajectory states and resolution operators.

3.3 Cross-Resolution Clean Latent Lifting

The Clean Latent Lifter (CLL) takes a 16-channel Wan VAE latent as input and produces an upscaled 16-channel latent, preserving the temporal dimension:

$$U_\psi^{\mathrm{CLL}} : \mathbb{R}^{B\times 16\times F\times h\times w} \to \mathbb{R}^{B\times 16\times F\times H\times W}.$$

The architecture adopts an encoder-resampler-decoder structure inspired by LTX-2. A 3D convolutional input layer projects the latents into a 256-dimensional hidden space, followed by 4 spatiotemporal residual blocks to extract rich spatiotemporal features. A spatial resampler then scales the resolution, after which 4 additional 3D residual blocks and a final 3D convolutional layer reconstruct the 16-channel output. This spatiotemporal processing enables the model to exploit information from adjacent latent frames while keeping the total frame count invariant.

We implement two distinct spatial resampling configurations. For the $480 \times 832 \to 720 \times 1248$ transition, the latent grid expands from 60×104 to 90×156. The network first inflates the channel dimension by a factor of 9, applies a $3\times$ spatial PixelShuffle, and finally performs a stride-2 downsampling using a fixed binomial blur kernel to achieve a rational $3/2\times$ upscaling:

$$60 \times 104 \xrightarrow{\times 3} 180 \times 312 \xrightarrow{/2} 90 \times 156.$$

For the $368 \times 640 \to 720 \times 1248$ transition, the latent grid expands from 46×80 to 90×156. Since the target size is not an exact integer multiple, the network executes a $2\times$ PixelShuffle to reach 92×160, followed by a center crop to isolate the exact 90×156 target grid.

The training objective for CLL is defined as:

$$\mathcal{L}_{\mathrm{CLL}} = \mathcal{L}_{\mathrm{rec}} + \lambda_{\mathrm{content}}\mathcal{L}_{\mathrm{content}} + \lambda_{\mathrm{temp}}\mathcal{L}_{\mathrm{temp}}.$$

where

$$\mathcal{L}_{\mathrm{rec}} = \mathbb{E}\left[\sqrt{(\widehat{z}_0^H - z_0^H)^2 + \epsilon^2}\right].$$

$$\mathcal{L}_{\mathrm{content}} = \|D(\widehat{z}_0^H) - z_0^L\|_1, \quad (1)$$

$$\mathcal{L}_{\mathrm{temp}} = \|\Delta_F \widehat{z}_0^H - \Delta_F z_0^H\|_1. \quad (2)$$

Here, D denotes a downscaling operator that maps the predicted high-resolution latent back to the low-resolution grid, and Δ_F represents the first-order temporal difference between adjacent latent frames. We set $\lambda_{\mathrm{content}} = 0.2$, $\lambda_{\mathrm{temp}} = 0.1$, and $\epsilon = 10^{-3}$ in our experiments. These three loss terms jointly constrain point-wise reconstruction, low-frequency content preservation, and temporal consistency. Because a low-resolution latent cannot uniquely determine

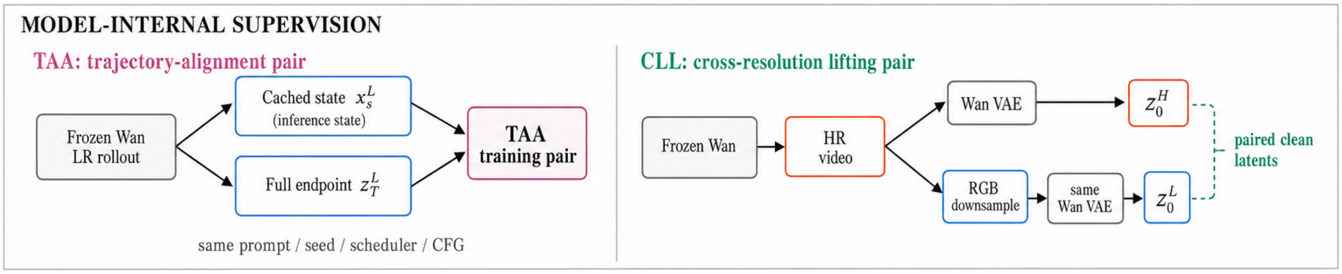

Figure 2: Model-endogenous supervision for TAA and CLL. TAA pairs the cached handoff state with the complete low-resolution trajectory endpoint under identical prompt, seed, scheduler, and CFG settings. CLL pairs clean latents encoded from a frozen-Wan-generated high-resolution video and its RGB-downsampled counterpart. Both training-pair streams are generated by the same frozen Wan model.

high-resolution details, CLL focuses on recovering a stable cross-scale representation, leaving the remaining high-frequency details to be synthesized and corrected by the subsequent HTR and high-resolution diffusion priors.

3.4 Handoff-Local Trajectory Alignment

TAA is implemented via LoRA (Hu et al. 2022) and is activated exclusively at the designated handoff step. The base Wan model remains completely frozen; low-rank updates of rank r are injected solely into the q, k, v, o projections of the attention modules and the $ffn.0, ffn.2$ linear layers of the Feed-Forward Networks (FFNs):

$$W' = W + \frac{\alpha}{r} BA.$$

Under the flow matching parameterization, the TAA-aligned clean latent prediction at step s is expressed as:

$$\tilde{z}_s^L = x_s^L - \sigma_s v_{\phi+\Delta\theta_s}(x_s^L, s, c).$$

The alignment target is the low-resolution trajectory endpoint z_T^L obtained by fully running the frozen model up to step T under identical conditions (text prompt, random seed, initial noise, and scheduler). The trajectory alignment loss is formulated as:

$$\mathcal{L}_{\text{align}} = \|\tilde{z}_s^L - z_T^L\|_1 + 0.1\|\tilde{z}_s^L - z_T^L\|_2^2.$$

For earlier handoffs (such as step 40), we introduce an additional temporal residual alignment loss with a weight of 0.05 to mitigate the propagation of frame-to-frame errors into the CLL module. While our framework also supports direct supervision via the velocity target $v^* = \frac{x_s^L - z_T^L}{\sigma_s}$, all experiments in this paper report results optimized using the clean latent alignment objective above.

By compressing the residual error of the low-resolution tail trajectory at the handoff step, TAA effectively projects the input of CLL back onto the clean latent training domain. While the system-level acceleration of TALH stems from the low-resolution prefix computations, TAA ensures trajectory consistency at fixed handoff points. Earlier handoffs, which inherently suffer from larger residual trajectory gaps, consequently yield a broader optimization space for TAA calibration.

3.5 Cached-Prefix State Consistency

Assuming TAA is activated strictly at the handoff step s, its LoRA parameters satisfy $\Delta\theta_k = 0$ for all steps $k < s$. Given a deterministic sampler alongside fixed text embeddings, random seeds, schedulers, and Classifier-Free Guidance (CFG) scales, the adapted generative process shares the identical transition functions as the base process prior to step s. Thus, by induction over the sampling steps, we establish:

$$x_k^{\text{adapted}} = x_k^{\text{base}}, \qquad \forall k \le s.$$

This property guarantees that the pre-cached base model prefix state x_s^L is mathematically identical to the state actually accessed by TAA during inference. For localized adaptations restricted to a single handoff step, caching the prefix state allows us to bypass redundant prefix computations entirely during inference without introducing any training-inference distribution drift, maintaining an $O(1)$ state retrieval overhead.

This consistency property relies on four strict conditions: TAA must be engaged exclusively at the handoff step; training and inference must share identical schedulers, CFG scales, and conditional inputs; the prefix sampling path must be entirely deterministic; and no unaligned stochastic operations can occur within the prefix trajectory. If TAA were extended across multiple steps, its parameters would alter subsequent states, requiring advanced training mechanisms such as student model rollouts, synchronized teacher predictions, or Exponential Moving Average (EMA) teacher models.

3.6 High-Resolution Trajectory Re-entry and Two Instances

Once TAA produces the aligned low-resolution clean latent, CLL outputs the upscaled high-resolution clean latent:

$$\tilde{z}_s^H = U_\psi^{\text{CLL}}(\tilde{z}_s^L).$$

The High-Resolution Trajectory Re-entry (HTR) phase then introduces target-resolution noise according to the schedule coefficient σ_{s+1} of the next timestep:

$$x_{s+1}^H = (1 - \sigma_{s+1})\tilde{z}_s^H + \sigma_{s+1}\epsilon^H.$$

The remaining denoising calculations from step $s+1$ to T are subsequently completed by the frozen Wan denoiser operating on the high-resolution latent space. Unlike an *Interpolated-Flow Re-entry* strategy that directly interpolates low-resolution velocity flows into the high-resolution space, TALH utilizes *Target-Resolution Noise Re-entry*. This design ensures that newly introduced spatial frequencies are synthesized natively by the high-resolution generative priors of the base model.

Our **50-step pipeline** uses Wan2.1 T2V-1.3B, configured with 50 inference steps, a sample shift of 8, and a CFG scale of 6. Extensive handoff-step scanning identified two representative operational points. The quality-centric **TALH-Q** (TALH@40) reserves 10 high-resolution steps, granting a larger budget for texture generation while managing a wider residual trajectory gap that requires substantial TAA alignment. Conversely, the efficiency-centric **TALH-E** (TALH@45) reserves only 5 high-resolution steps to minimize latency; its handoff state sits closer to a fully converged clean latent, resulting in more stable CLL inputs and narrower TAA corrections. Together, these two instances define the quality-efficiency Pareto frontier of TALH.

Our **4-step pipeline** leverages Wan2.1 T2V-14B StepDistill-CfgDistill, operating on four nominal timesteps $[1000, 750, 500, 250]$ with a sample shift of 5. The distilled instance **TALH-D4** (TALH@3-of-4) runs low-resolution sampling for the first two steps. At step 3, TAA predicts the aligned low-resolution clean latent, which is scaled by CLL and injected back into the high-resolution trajectory via HTR at step 4. A single final high-resolution denoising step completes the generation.

4 Experiments

4.1 Experimental Objectives

Our evaluation verifies TALH across both system-level and module-level dimensions. System-level experiments benchmark the end-to-end latency and generative quality of TALH-Q and TALH-E against Native-HR Sampling, while validating compatibility with few-step samplers via the 4-step distilled model. Module-level experiments independently isolate the cross-resolution lifting performance of CLL against fixed trilinear interpolation, quantify TAA's calibration of the residual trajectory gap, and dissect their main effects and interactions through a factorial framework. Finally, handoff-step scanning illuminates the tight coupling between low-resolution trajectory errors, high-resolution refinement budgets, and overall inference speedups.

4.2 Data and Training Setup

The 50-step pipeline utilizes approximately 1,000 high-resolution reference videos generated by Wan2.1 T2V-1.3B, each spanning 81 frames at 720×1248 resolution and a framerate of 16 fps. Bicubic interpolation produces the corresponding 480×832 and 368×640 downscaled counterparts, which are subsequently processed through the Wan VAE to yield aligned cross-resolution lifting pairs. Trajectory alignment pairs are constructed by extracting the low-resolution intermediate states immediately preceding the target handoff step alongside the full low-resolution trajectory endpoints of the same samples.

The 4-step pipeline utilizes roughly 5,000 videos generated by Wan2.1 T2V-14B StepDistill-CfgDistill. The CLL module is trained on $368 \times 640 \rightarrow 720 \times 1248$ clean latent pairs, while TAA is trained on the low-resolution states preceding step 3 paired with the complete step 4 low-resolution endpoints. All training, validation, and testing partitions are strictly separated by unique text prompts and random seeds to eliminate data leakage.

CLL is optimized using AdamW with a learning rate of 1×10^{-4} for up to 50k steps, using $bf16$ precision and an EMA decay rate of 0.9999. TAA is optimized via AdamW with a learning rate of 5×10^{-5} for up to 10k steps, employing a LoRA configuration of rank$/\alpha = 32/32$. Training is conducted across 4 NVIDIA H100 GPUs, with CLL and TAA requiring approximately 8 and 33 hours, respectively. To ensure clean measurements, all efficiency metrics are averaged over 5 independent cold-start runs.

Importantly, because all training videos are produced entirely by the frozen generator followed by spatial downscaling, this setup establishes a model-specific, generative latent upscaling task. TALH effectively distills Wan's intrinsic high-resolution generative capabilities into lightweight spatial and trajectory adapters without requiring any external paired datasets.

4.3 Baselines and Factorial Experiments

We evaluate TALH against the following baseline configurations, including JTSL for qualitative analysis of joint regression characteristics:

1. **Native-HR Sampling:** Standard high-resolution inference using the original model, where all steps are executed natively on the 720p latent space.
2. **Native-LR Sampling:** Standard low-resolution inference across all steps followed by direct VAE decoding.
3. **Trilinear Handoff:** Mixed-resolution sampling without trajectory alignment, using LightX2V-style trilinear upscaling at the handoff step.
4. **CLL-only Handoff:** Trajectory alignment is disabled, replacing trilinear interpolation solely with our CLL module.
5. **TAA + Trilinear Handoff:** Engages TAA for trajectory alignment but falls back to trilinear interpolation for upscaling.
6. **TALH:** Our full framework combining the TAA-aligned trajectory state with the CLL resolution operator.
7. **Oracle-Aligned CLL:** Bypasses TAA by directly feeding the true low-resolution trajectory endpoint into CLL, serving as the theoretical upper bound for trajectory alignment.
8. **Joint Trajectory-Scale Lifting (JTSL):** A baseline model trained to directly map the low-resolution intermediate prediction at the handoff step to the high-resolution target latent in a single step.

9. **Endpoint Re-entry Baseline:** Runs the low-resolution trajectory to completion, applies CLL upscaling, and performs a single low-intensity high-resolution refinement step via HTR.

Our core evaluations adopt a 2×2 factorial design at steps 40 and 45, crossing the trajectory states $\{$Unaligned, TAA-aligned$\}$ with the resolution operators $\{$Trilinear, CLL$\}$. Fixing the resolution operator isolates the main effect of TAA, fixing the switching state isolates the main effect of CLL, and evaluating all four combinations maps their explicit interactions. Qualitative profiles of JTSL are leveraged to analyze high-frequency damping induced by joint trajectory-scale regression.

4.4 Metrics and Fairness

For trajectory alignment, we report latent L_1/MSE relative to the full low-resolution endpoint, temporal L_1 across adjacent frames, decoded LPIPS (Zhang et al. 2018), PSNR/SSIM, and sample-wise win rates. For CLL, we report clean latent Charbonnier/L_1 errors, decoded PSNR/SSIM/LPIPS, temporal difference errors, and high-frequency energy discrepancies. End-to-end video generation quality is measured via VBench (Huang et al. 2024) alongside human evaluation to assess subject consistency, temporal stability, and perceptual quality.

The comprehensive VBench score (**VBench-5**) is defined as the unweighted mean of five core dimensions: *subject consistency*, *background consistency*, *motion smoothness*, *aesthetic quality*, and *imaging quality*. To guarantee strict equity, all pairwise configurations share identical text prompts, random seeds, initial noise, schedulers, and guidance parameters. Prompt-level score deltas are evaluated via 10,000 paired bootstrap iterations to compute 95% confidence intervals, and two-sided exact sign tests measure win-loss significance.

An anonymous human study evaluates fine details, artifacts, temporal stability, structural/identity preservation, and overall quality. Each text prompt is rated by 3 independent judges under a randomized double-blind presentation, with the majority vote across 10 prompts serving as the statistical unit. Inter-rater reliability is verified using observed agreement and Fleiss' Kappa.

Efficiency evaluations report end-to-end cold-start latency per video, peak GPU memory, the number of low- versus high-resolution denoising evaluations, and the speedup ratio relative to Native-HR Sampling. Timings include all processing overheads from TAA, CLL, HTR, VAE operations, and dynamic model swapping. Quality benchmarks are computed on the same 10 matched prompts.

5 Results and Analysis

5.1 Core Comparison: Quality-Efficiency Pareto Frontier Against Native-HR Sampling

Table 1 evaluates the standard 50-step Native-HR Sampling against our two primary TALH operational points. Native-HR Sampling exhibits an average runtime of 253.10 seconds. In comparison, TALH-Q (handoff at step 40) and TALH-E (handoff at step 45) reduce this latency to 138.36 seconds and 114.26 seconds, yielding latency reductions of 45.33% and 54.85%, respectively. They preserve 97.76% and 97.53% of Native-HR's generative quality on VBench-5. Because TALH leaves the total number of denoising iterations unchanged, its system acceleration is derived entirely from offloading 40 or 45 steps to the computationally efficient low-resolution latent grid. Figure 3(a) illustrates the quality-latency trade-offs of these configurations, and Figure 3(b) decomposes the dimension-wise VBench deviations relative to Native-HR Sampling.

Compared to Native-HR Sampling, TALH-Q and TALH-E experience minor VBench-5 concessions of only 2.24% and 2.47% while unlocking substantial $1.83\times$ and $2.22\times$ speedups. TALH-Q leverages its extended high-resolution suffix to achieve higher overall generation fidelity, whereas TALH-E pushes online inference throughput to its practical limits, successfully mapping out a highly controllable quality-efficiency Pareto frontier.

5.2 Clean Latent Lifting Outperforms Fixed Interpolation

Figure 4(a) and Table 2 evaluate our CLL operator against classical trilinear upscaling across 50 paired clean latent samples. For the 480p $\rightarrow$ 720p transition, CLL dramatically compresses the latent L_1 error from 0.1929 to 0.0991 (a 48.6% relative reduction). Simultaneously, decoded LPIPS drops from 0.2358 to 0.0629, PSNR improves by 5.71 dB, and temporal L_1 variation decreases by 30.8%. Except for a single sample on SSIM (49/50), CLL secures comprehensive wins across all 50 test samples for PSNR, LPIPS, and temporal L_1 consistency.

In the more aggressive 368p $\rightarrow$ 720p setting, CLL reduces latent L_1, LPIPS, and temporal L_1 errors by 33.4%, 51.1%, and 15.5%, respectively, demonstrating uniform improvements across visual quality, structure, and high-frequency energy profiles. These metrics confirm that CLL successfully models an authentic cross-scale mapping that tightly matches the high-resolution VAE manifold, moving far beyond superficial image sharpening.

5.3 Trajectory Alignment Narrows Residual Endpoint Gaps

Figure 4(b) and Table 3 detail the performance of TAA operating at an alignment intensity of 0.75. For the step 40 handoff, TAA drives the residual endpoint L_1 error down from 0.03215 to 0.02385 (a 25.82% relative drop). For the step 45 handoff, it achieves a 21.03% reduction from 0.02363 to 0.01866. Both configurations register lower errors across all 10 evaluation samples (10/0 win-loss ratio), with 95% bootstrap confidence intervals strictly excluding zero, evaluated as statistically significant via a two-sided exact sign test ($p = 0.00195$). This calibration translates to a 2.55 dB and 2.23 dB boost in decoded PSNR alongside a 23.44% and 21.24% reduction in latent temporal L_1 variations. Notably, the absolute L_1 error improvement at step 40 (0.00830) visibly exceeds that of step 45 (0.00497), verifying that earlier handoff points offer an expanded optimization window for trajectory alignment.

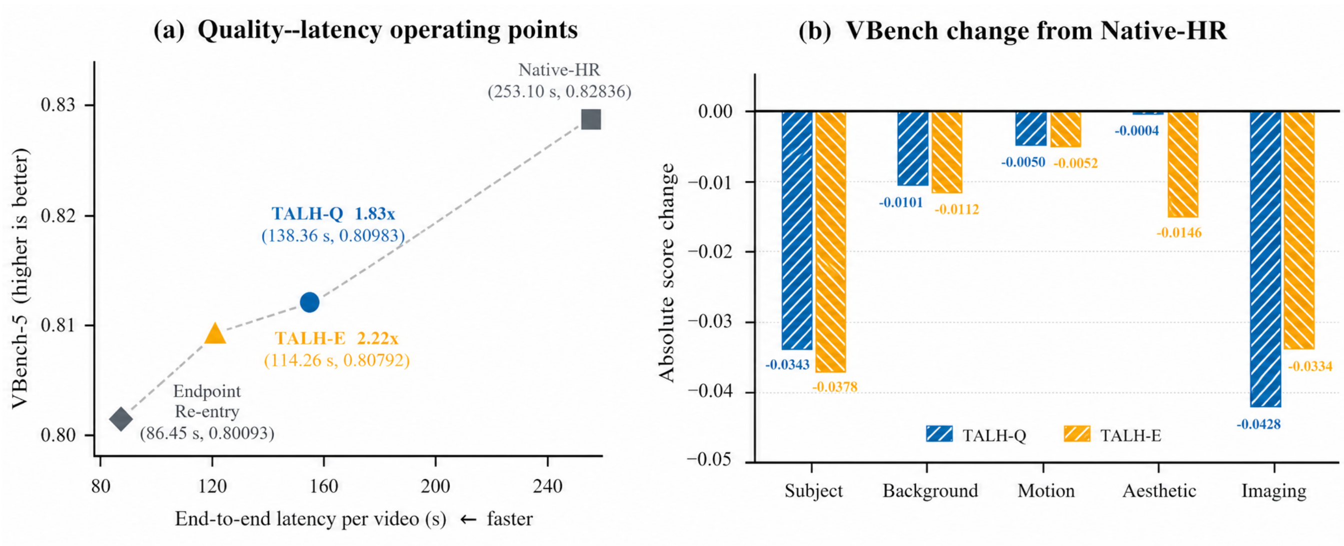

Figure 3: Quality-efficiency trade-off relative to Native-HR Sampling. (a) TALH-Q and TALH-E provide quality- and efficiency-oriented operating points; Endpoint Re-entry is faster but incurs a larger quality decrease. (b) Absolute per-dimension VBench changes relative to Native-HR.

Method	LR/HR evals.	Time (s) ↓	Latency red.	Speedup ↑	VBench-5 ↑	Retention ↑
Native-HR Sampling	0/50	253.10 ± 1.53	—	1.00×	**0.82836**	100.00%
TALH-Q (Step 40)	40/10	138.36 ± 0.50	45.33%	1.83×	0.80983	97.76%
TALH-E (Step 45)	45/5	$\mathbf{114.26 \pm 1.52}$	**54.85%**	**2.22**×	0.80792	97.53%

Table 1: Core system comparison against Native-HR Sampling. Timings average five cold-start runs; VBench-5 uses ten matched prompts. All TAA, CLL, HTR, VAE, and dynamic-loading overheads are included.

Intensity scanning reveals a non-monotonic relationship between raw alignment magnitude and downstream generation quality. At step 40, alignment intensities of 0.5 and 1.0 reduce latent L_1 error by only 0.00012 and 0.00071, respectively. An intermediate intensity of 0.75 achieves the lowest L_1 error, a perfect $10/0$ sample-wise win rate, and the highest downstream VBench-5 score. Modulating the alignment strength in this manner ensures that the calibrated states remain well-conditioned for the CLL and HTR pipelines.

5.4 Factorial Analysis: CLL Drives Primary Generation Gains

Figure 4(c) and Table 4 break down the complete $\{$Unaligned, TAA-aligned$\} \times \{$Trilinear, CLL$\}$ factorial experiment. Integrating CLL yields substantial VBench-5 improvements of +0.03085, +0.04066, and +0.03874 at step 40, step 45, and on the 4-step sampler, respectively. Relative to the baseline Trilinear Handoff, the full TALH framework secures total gains of +0.03171, +0.04016, and +0.03884. Prompt-level statistics demonstrate that CLL-driven improvements are concentrated in *motion smoothness*, *aesthetic quality*, and *imaging quality*, with TALH-Q logging win/loss tallies of $10/0$, $9/1$, and $9/1$, and TALH-E returning $10/0$, $9/1$, and $10/0$, respectively.

The downstream impact of TAA is more structured at the dimension level than its small change in aggregated VBench-5 suggests. With CLL fixed, enabling TAA consistently increases the *imaging quality* component by +0.00599, +0.00254, and +0.00479 at step 40, step 45, and on the 4-step sampler, respectively. These gains are partly offset by changes in aesthetic quality and consistency, yielding aggregate VBench-5 changes of +0.00087, -0.00050, and +0.00010. Together with the endpoint results in Table 3, this pattern characterizes TAA as a detail-oriented trajectory correction that selectively recovers imaging quality lost during resolution handoff, rather than a uniform enhancer of every automatic quality dimension.

5.5 Human Evaluation Confirms Detail Recovery and Its Temporal Trade-off

Human evaluation tests whether the selective imaging-quality recovery identified by VBench is perceptually visible. Conditioned on the same CLL module within TALH-E, enabling TAA yields win/loss/tie tallies of $9/0/1$ for fine details and $6/0/4$ for overall quality, while temporal stability favors the unaligned state at $0/8/2$. TALH-Q shows the same direction, with $8/2/0$ for fine details, $6/2/2$ for overall quality, and $1/9/0$ for temporal stability. Raters therefore perceive TAA outputs as containing finer localized structures and often prefer their overall rendering, even though the additional high-frequency detail introduces a measurable temporal trade-off. This paired perceptual evidence com-

Input → Output	Operator	Latent L_1 ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Temp. L_1 ↓
480p → 720p	Trilinear	0.1929	24.29	0.7159	0.2358	0.02121
480p → 720p	CLL	**0.0991**	**30.00**	**0.8696**	**0.0629**	**0.01468**
368p → 720p	Trilinear	0.2444	23.42	0.6740	0.3352	0.02388
368p → 720p	CLL	**0.1628**	**24.10**	**0.7592**	**0.1640**	**0.02017**

Table 2: Operator-level comparison between CLL and trilinear upscaling ($n = 50$). Each row uses the matched high-resolution VAE latent as reference.

Trajectory / handoff	Unaligned L_1 ↓	TAA L_1 ↓	Reduction	95% CI	Win/Loss	p
Wan50, Steps 40 → 50	0.03215	**0.02385**	25.82%	[0.00479, 0.01302]	10/0	0.00195
Wan50, Steps 45 → 50	0.02363	**0.01866**	21.03%	[0.00260, 0.00840]	10/0	0.00195
Distill4, Steps 3 → 4	0.04286	**0.04070**	5.03%	[0.00148, 0.00299]	10/0	0.00195

Table 3: Paired trajectory-alignment results. The first two rows use native 368p trajectories; the third uses an independent Distill4 transfer set.

plements the nearly unchanged aggregate VBench-5 score, whose five-way average dilutes improvements concentrated in imaging detail.

CLL remains the primary source of broad perceptual improvement: for the step 45 handoff, both CLL-only Handoff and the full TALH-E framework secure perfect $10/10$ prompt-level win rates across artifacts, details, overall quality, and structural/identity preservation relative to Trilinear Handoff. TAA contributes a narrower but perceptually visible refinement of local detail on top of CLL. This division of labor is consistent with the VBench-5 factorial findings. Inter-rater agreement statistics are provided in the Supplementary Material.

5.6 Evaluation Against Strong Speed Baselines and Handoff Planning

The Endpoint Re-entry Baseline—which runs the low-resolution trajectory to completion, applies CLL upscaling, and performs a single high-resolution refinement step—requires 86.45 $\pm$ 2.87 seconds, reaching a 2.93$\times$ speedup. However, it suffers a substantial 3.31% decline in VBench-5 (0.80093) relative to Native-HR Sampling. Paired bootstrap analysis yields a 95% confidence interval of $[-0.05128, -0.00577]$ for the VBench-5 delta, with inferior scores across $9/10$ prompts ($p = 0.02148$). While a single refinement step maximizes inference speed, its overall generative quality falls short of TALH-Q and TALH-E.

TALH-Q adds 24.09 seconds of latency compared to TALH-E but recaptures a +0.00191 VBench-5 gain. Dimension-wise breakdowns show that TALH-Q improves subject consistency and aesthetic quality by 0.00354 and 0.01419, respectively, while TALH-E improves imaging quality by 0.00939. This directly aligns with our handoff-step scanning profiles: an extended high-resolution suffix excels at structural and texture synthesis, whereas delaying the handoff step directly minimizes runtime. Peak memory utilization remains stable at approximately 26 GiB across Native-HR, both TALH variants, and the Endpoint Re-entry Baseline, confirming that TALH's system advantages are concentrated entirely in inference speedups.

5.7 Transferability to Distilled Samplers

TALH is highly compatible with temporal step distillation frameworks. In transfer testing using the 368p trajectory on the 4-step distilled model, TAA reduces endpoint L_1 and LPIPS errors by 5.03% and 5.65%, respectively, achieving consistent error reductions across all 10 test samples. In end-to-end evaluations, Trilinear Handoff registers a VBench-5 score of 0.81796, which TALH-D4 elevates to 0.85680. Crucially, introducing TAA and CLL only marginally shifts cold-start latency from 167.09 seconds to 170.29 seconds. These results demonstrate that TALH's cross-resolution lifting mechanism stacks cleanly with few-step samplers, where CLL delivers the primary quality improvements and TAA performs localized trajectory alignment.

Furthermore, handoff-step scanning reveals that JTSL introduces severe high-frequency damping when forced to simultaneously model spatiotemporal trajectory restoration, cross-scale geometric mapping, and texture inference. By decoupling trajectory alignment from clean latent lifting, TALH mitigates this epistemic uncertainty, allowing CLL to preserve crisp, high-fidelity spatial details.

5.8 Qualitative Comparison

Figure 1(a) presents a spatial comparison of trilinear upscaling, CLL-only Handoff, and the complete TALH-Q framework on content-aligned video frames at step 40. The comparison highlights CLL's reconstruction of high-frequency textures and TAA's localized adjustments. Figure S1 in the Supplementary Material provides an extended dual-crop spatial panel and evaluates temporal sequences at step 45 to analyze structural fidelity and texture attachment for TALH-E.

The `Native-HR (estimated)` baseline represents a full 368p sampling path sharing identical content and motion characteristics, upscaled purely for structural reference. Because native 720p inference can drift into divergent generative contents even under identical random seeds, true 720p Native-HR Sampling is reserved for the quantitative metrics in Table 1 rather than frame-by-frame visual pairings.

Sampler / handoff	Trilinear	TAA+Tri.	CLL-only	TALH	CLL gain	TAA gain	Total gain
Wan50 / Step 40 (TALH-Q)	0.77812	0.77901	0.80896	**0.80983**	+0.03085	+0.00087	+0.03171
Wan50 / Step 45 (TALH-E)	0.76776	0.76734	**0.80842**	0.80792	+0.04066	-0.00050	+0.04016
Distill4 / Step 3-of-4	0.81796	0.81909	0.85670	**0.85680**	+0.03874	+0.00010	+0.03884

Table 4: End-to-end VBench-5 factorial results ($n = 10$). "TAA gain" is the marginal change from enabling TAA while keeping CLL fixed.

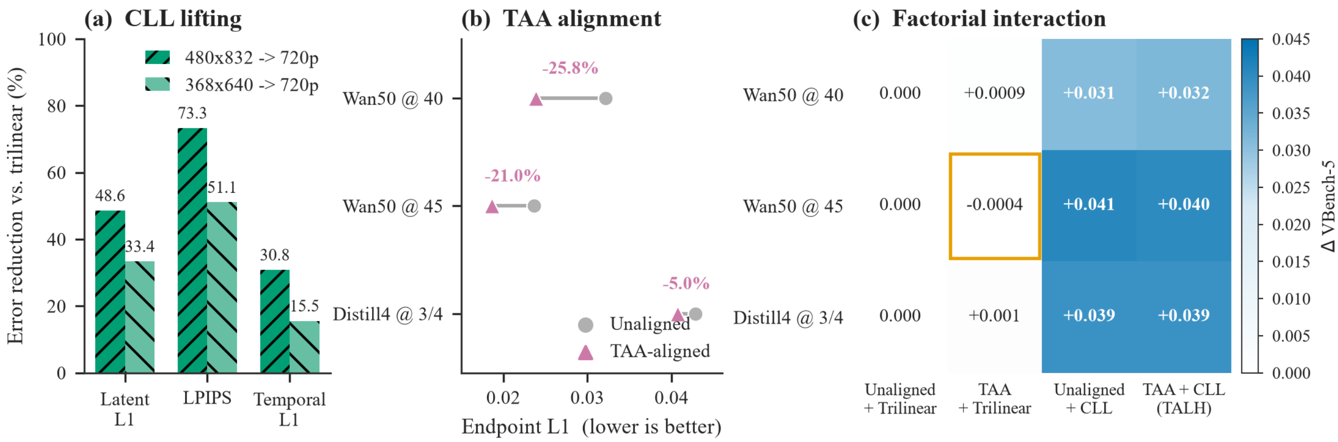

Figure 4: Module effects and interactions in TALH. (a) CLL reduces latent, perceptual, and temporal errors relative to trilinear interpolation. (b) TAA reduces the low-resolution trajectory-endpoint gap on Wan50 and the 4-step distilled model. (c) The 2×2 factorial results show that CLL provides the dominant VBench-5 gain, while TAA performs local trajectory correction.

6 Discussion and Limitations

TALH introduces a clear division of labor: it offloads timestep- and scheduler-dependent trajectory errors to TAA, isolates CLL training within a stable clean latent domain, and utilizes HTR to leverage the base model's high-resolution diffusion priors for high-frequency detail generation. This structured decomposition creates an interpretable pipeline from modular objectives to end-to-end generation quality. The endpoint metrics, consistently positive imaging-quality deltas, and human detail preferences jointly show that TAA selectively restores local spatial detail. This benefit does not translate linearly into aggregated VBench-5 because the five-way average includes largely unchanged dimensions and because the added high-frequency detail trades off against aesthetic and temporal stability. TAA's final effect also remains tied to the input distribution of CLL and the length of the high-resolution suffix. Consequently, the 2×2 factorial design and paired human evaluation are both essential for resolving these module interactions.

Crucially, TAA and CLL operate within a model-endogenous self-distillation framework. The frozen Wan model concurrently provides high-resolution reference videos for cross-resolution pairs and full low-resolution trajectories for trajectory alignment pairs. This allows both modules to learn how to efficiently repurpose the base model's internal generative priors, eliminating the need for external paired datasets, independent upscalers, or heavy teacher architectures.

Despite these advantages, TALH has several limitations. First, because CLL specifically maps the interaction between the Wan VAE, the model's generative distribution, and spatial downscaling, its transferability across different VAE architectures and alternative base generators remains unverified. Second, a slight distribution gap persists between the downsampled clean latents used to train CLL and the native low-resolution trajectory endpoints targeted by TAA. Third, because fixed handoff steps map to specific TAA parameter states, implementing multi-step or adaptive handoffs will require modeling explicit timestep conditioning. Finally, our training pipeline focuses exclusively on spatial scaling; while highly effective for transitions within generation trajectories, these modules are not designed to handle real-world video degradation like compression artifacts, motion blur, or sensor noise.

From an application perspective, mixed-resolution sampling significantly lowers the computational and energy barriers for high-resolution video generation, accelerating the scalability of synthetic media production. The downstream content risks of TALH are entirely inherited from the base video diffusion model. Deployments should strictly adhere to the base model's licensing agreements, apply explicit synthetic media watermarking, and maintain identical safety filtering protocols.

7 Conclusion

This paper presents TALH, a lightweight latent-space framework designed to accelerate high-resolution video generation by dynamically pairing a low-resolution prefix trajec-

Comparison (former vs. latter)	Details (W/L/T)	Overall (W/L/T)	Temporal (W/L/T)
TAA vs. Unaligned, both CLL (Step 40)	8/2/0	6/2/2	1/9/0
TAA vs. Unaligned, both CLL (Step 45)	**9/0/1**	**6/0/4**	0/8/2
CLL-only vs. Trilinear (Step 45)	**10/0/0**	**10/0/0**	**8/0/2**
TALH-E vs. Trilinear (Step 45)	**10/0/0**	**10/0/0**	**9/0/1**

Table 5: Prompt-level majority votes ($n = 10$, three raters per prompt). Bold entries have two-sided exact sign-test $p < 0.05$.

tory with a high-resolution sampling suffix. TALH resolves the resolution transition by decomposing it into three explicit phases: TAA for residual trajectory alignment, CLL for cross-resolution clean latent lifting, and HTR for high-resolution trajectory re-entry. These modules are trained via a self-distillation paradigm using data generated entirely by the frozen base model.

Our experiments demonstrate that TALH-Q and TALH-E deliver substantial $1.83\times$ and $2.22\times$ speedups over Native-HR Sampling while preserving 97.76% and 97.53% of generative quality on VBench-5. CLL consistently outperforms standard trilinear interpolation across multiple scaling ratios, while TAA reliably narrows residual trajectory gaps in both standard 50-step and advanced 4-step distilled models. These results confirm the effectiveness of separating trajectory alignment from latent lifting, providing a highly controllable, lightweight, and scalable quality-efficiency trade-off for online high-resolution video generation.

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